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**Design and Development of an Algorithm for Object Shape Detection Based on
Optoelectronic Technology**

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Declaration

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ABSTRACT

This dissertation presents the design and development of an algorithm for object shape detection based on optoelectronic technology, specifically utilizing a 4-channel Time-of-Flight (ToF) sensor array. In the context of Industry 4.0 automation, traditional computation vision-based systems provide high-resolution shape recognition but demand substantial computational resources that are often unsuitable for standard edge microcontrollers. Conversely, conventional low-cost alternatives, such as ultrasonic and single-point infrared (IR) sensors, suffer from acoustic crosstalk and target reflectance bias, respectively.

To bridge this technological gap, this study integrates four VL6180 optoelectronic ToF sensors and introduces a novel mathematical algorithm that maps raw multi-point distance data into an 8-bit grayscale occupancy grid (0-255). This lightweight data representation significantly reduces computational overhead, enabling real-time object shape detection and profile estimation on resource-constrained embedded systems. The system's performance is rigorously benchmarked on a stepper-motor-driven linear guide, evaluating its algorithm accuracy, angular resolution, and robustness against varying surface reflectance compared to conventional sensing methods. The results demonstrate that the 8-bit 4-ToF algorithmic array provides a highly pragmatic, color-independent solution for object shape recognition in industrial environments.

Keywords: Object Shape Detection; Optoelectronic Technology; Time-of-Flight (ToF) Sensor Array; 8-Bit Grayscale Mapping; Edge Computing; Industrial Automation; Algorithm Development

1. $\emptyset \nearrow \nearrow \odot \square \cup \text{---} \parallel \nearrow \nearrow \equiv \square \emptyset$

1.1 $A \equiv (M)$

The primary aim of this research is the design and development of an algorithm for object shape detection based on optoelectronic technology. By integrating a multi-node Time-of-Flight (ToF) sensor array and converting linear distance measurements into a 8-bit grayscale depth map, the system seeks to provide accurate, real-time object profiling with minimal computational overhead, specifically optimized for edge-node industrial applications.

1.2 В $\angle$ $\square$ $\cap$ $\cup$ $\odot$ $\ominus$ $\square$ $-$ $\emptyset$ $\bigcirc$

In the era of Industry 4.0 [1], intelligent sensors are fundamental to the operation of automated systems, collaborative robots (Cobots), and autonomous guided vehicles (AGVs) [2] [18]. These systems rely on continuous, real-time data processing to safely interact with their environments and human operators. Historically, precise spatial measurement required complex and labor-intensive mechanical tools. Modern digital methods utilize optoelectronic technologies to achieve precise non-contact estimation and shape recognition.

Currently, the industry faces a dichotomy in sensing technology. High-end systems, such as

LiDAR and stereo vision cameras, offer comprehensive 3D spatial data but require high capital investment and extensive datasets for processing. On the other end, low-cost sensors such as ultrasonic and single-point infrared (IR) are widely deployed but provide limited spatial coverage, making them inadequate for complex shape analysis. There is an increasing demand for an intermediate algorithm solution-a sensor architecture that delivers multi-point shape detection without the extreme computational burden of high-resolution image processing [17] [20].



Accurate real-time object shape detection is critical for industrial and automation. However, relying on single-point optoelectronic sensors restricts the system to one-dimensional distance ranging, rendering it incapable of detecting an object's width, contour, or lateral trajectory.

When attempting to scale low-cost sensors into multi-node array to expand spatial awareness, inherent physical limitations emerge. Ultrasonic sensors possess wide angular sensitivity lobes; thus, deploying multiple ultrasonic sensors in proximity results in severe acoustic interference and echo confusion [4], [5]. Similarly, conventional IR sensors are biased by varying reflectance properties of the target surface [3]. Therefore, the inability of conventional low-cost to reliably and efficiently map multiple spatial points highlights a significant gap in current shape detection algorithms.



To address the defined problem and achieve the overarching aim, the following specific objectives have been established.

Hardware Integration: To design and construct an optoelectronic sensing array by integrating four VL6180 ToF sensors via dynamic I2C

address assignment with an Arduino Uno Mega, ensuring synchronous multi-sensor bus communication across an expanded Field of View.

Algorithm Development: To formulate and implement a normalization algorithm that translates physical distance reading from the 4-ToF array into an 8-bit grayscale intensive matrix (0-255) for shape recognition.

Experimental Setup and Benchmarking: To deploy the optoelectronic array on a precisely controlled, stepper-motor-driven linear guide to evaluate the algorithm's accuracy and response time against varying target geometries and surface colors.

Validation and Verification: To critically compare the system's shape detection performance-specifically regarding multi-sensor crosstalk and color independence-against conventional ultrasonic and IR ranging techniques.



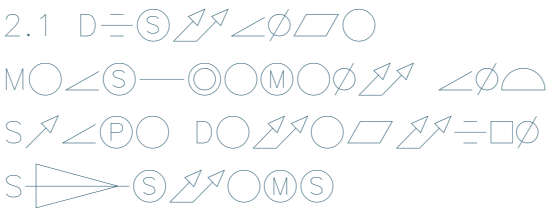
The development of an object shape detection algorithm utilizing a 4-channel optoelectronic Time-of-Flight (ToF) sensor array holds substantial potential for advancing industrial automation and robotics. As industry increasingly shifts towards the deployment of autonomous guided vehicles (AGVs) and collaborative robots, the demand for affordable, real-time spatial awareness systems has surged. By introducing a mathematical normalization algorithm that maps raw multi-point distance data into an 8-bit grayscale occupancy grid, this research provides a highly efficient method for object profiling that circumvents the immense computational overhead typically associated with high-resolution stereo vision cameras.

Furthermore, this study addresses the critical physical limitations of conventional low-cost sensors. By utilizing ToF technology, the proposed algorithm system eliminates the severe

acoustic crosstalk inherent in ultrasonic arrays and the target reflectance biases common in infrared sensors. The significance of this research lies in its pragmatic approach: delivering a cost-effective, color-independent, and computationally lightweight framework that enables resource-constrained edge microcontrollers to perform reliable shape detection in dynamic industrial environments.

2.LITERATURE REVIEW

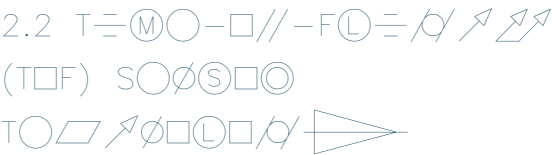
This chapter critically reviews the existing literature surrounding distance measurement technologies and object shape detection algorithms. This review is structured to first evaluate the broader landscape of sensor technologies, before focusing on Time-of-Flight (ToF) principles, and finally examining the mathematical model algorithms necessary for surface reconstruction.



Distance measurement and spatial profiling are foundational to modern industrial automation, collaborative robotics (Cobots), and autonomous guided (AGVs). The transition from Industry 1.0 to Industry 4.0 has necessitated the shift from manual mechanical measurement tools to intelligent, non-contact digital sensors capable of real-time data acquisition.

In the realm of autonomous systems, distance measurement techniques are broadly categorized into active and passive methods, predominantly relying on camera-based systems such as mono and stereo vision, utilizing computer vision algorithms to estimate depth and recognize objects. While highly accurate, these systems demand substantial computational resources and extensive training datasets, which are often unfeasible for low-cost edge

microcontrollers. Conversely, active methods emit signals such as sound, light or radio waves and measure the reflection to determine proximity. Traditional active their low cost and response times. However, their application in multi-node arrays for angular sensitivity lobes, leading to sever acoustic echo confusion when operated in proximity. Similarly, conventional IR sensors suffer from non-lineal behavior and are highly sensitive to the reflectance and color of the target surface, resulting in distorted environmental maps.



To overcome the limitations of conversional IR and ultrasonic sensors, Time-of-Flight (ToF) technology has emerged as a robust solution for precise spatial measurement. ToF sensors, including Light Detection and Ranging (LiDAR) systems, operate by emitting modulated infrared light pulse and calculating the distance based on the time taken for the photons to reflect off an object and return to the receiver [7], [8], [19].

Unlike traditional IR triangulation sensors, optoelectronic ToF sensors measure the absolute distance independent of the target's surface color or reflectance. Solid-state ToF sensors enable real-time 3D scene analysis with minimal mechanical complexity. However, the resolution of these sensors can be constrained by shot noise, which arises from the discrete nature of photons.

The VL6180 sensor, developed by STMicroelectronics, is a prominent example of an integrated ToF module [9]. It combines an IR emitter, a range sensor, and ambient light sensor within a single compact package, utilizing proprietary FlightSense technology, the VL6180 accurately measures proximity up to 400 mm, providing a digital output via an I2C interface. The crucial nature of the emitted laser pulse, which significantly mitigates the inter-channel

crosstalk that plagues ultrasonic equivalents. This makes the VL6180 an ideal candidate for constructing a multi-node optoelectronic array for shape profiling.

2.3 $M \angle \nearrow \nearrow \circ \textcircled{M} \angle \nearrow \nearrow = \square \angle \textcircled{L}$
 $A \textcircled{L} \nearrow \square \textcircled{\circ} = \nearrow \nearrow \textcircled{M} \textcircled{S} // \square \textcircled{\circ}$
 $S - \textcircled{\circ} // \angle \square \textcircled{\circ}$
 $R \textcircled{\circ} \square \square \emptyset \textcircled{S} \nearrow \textcircled{\circ} - \square \nearrow \nearrow = \square \emptyset$

The acquisition of raw distance data from a sensor array is insufficient for object recognition without the application of a robust normalization algorithm. In multi-axis scanning mechanisms, the physical geometry of the sensor arrangement must be mathematically modelled to translate linear distance reading into Cartesian spatial coordinates (x, y) .

This transformation typically involves trigonometric functions that account for the sensor's mounting angle and its displacement from a central original point. Once the raw distances are converted into a two-dimensional spatial coordinate system, surface profiling algorithms can be applied. While simple linear interpolation (point-to-point connection) provides a rudimentary outline, it fails to account for sensor noise and lacks the analytical capability to classify geometric shapes.

Advanced surface reconstruction relies on polynomial curve fitting (regression analysis). By processing the transformed coordinates through a quadratic formulation ($y = ax^2 + bx + c$), the algorithm can establish a best-fit continuous profile of the target. In this mathematical context, the coefficients serve as classifiers; for instance, the sign of the leading coefficient (a) determines the concavity of the profile, enabling the system to autonomously distinguish between flat, convex, and concave object surfaces. The integration of such algorithms allows resource-constrained embedded systems to perform complex shape detection without the need for high-resolution image processing.

3. METHODOLOGY

This chapter details the experimental setup, hardware configuration, and the algorithmic procedures employed to develop the multi-node optoelectronic shape detection system. The methodology is systematically divided into hardware integration, firmware architecture for sensor communication, data acquisition, and mathematical surface reconstruction.

3.1 $H \angle \textcircled{\circ} \cap \vee \angle \textcircled{\circ} \textcircled{\circ}$
 $I \emptyset \nearrow \textcircled{\circ} \nearrow \textcircled{\circ} \angle \nearrow \nearrow = \square \emptyset \angle \emptyset \cap$
 $S \textcircled{\circ} \emptyset \textcircled{S} \square \textcircled{\circ}$
 $\square \square \emptyset // = \nearrow - \textcircled{\circ} \angle \nearrow \nearrow = \square \emptyset$

To achieve accurate multi-point spatial, a 4-channel sensor array was constructed utilizing VL6180 Time-of-Flight (ToF) optoelectronic sensors. The sensors were mounted on bespoke 3D-printed fixture, arranged in a precise geometric configuration to establish a convergent field of view targeting the object. Servo motors were utilized to securely position the lateral and central sensors at specific angles (θ) relative to the baseline origin.

The central processing unit driving the system is an Arduino Mega 2560 microcontroller [11], [12]. This board was specifically selected for its abundant General-Purpose Input/Output (GPIO) pins, which are essential for controlling the individual hardware states of the sensor array, alongside its robust I2C communication capabilities.

3.2 $M = \square \textcircled{\circ} \square \square \emptyset \nearrow \textcircled{\circ} \square \textcircled{L} \textcircled{L} \textcircled{\circ} \textcircled{\circ}$
 $F = \textcircled{\circ} \textcircled{M} \vee \angle \textcircled{\circ} \textcircled{\circ} \angle \emptyset \cap$
 $D \rightarrow \emptyset \angle \textcircled{M} = \square \text{ I2C}$
 $A \textcircled{S} \textcircled{S} = \nearrow \emptyset \textcircled{M} \textcircled{\circ} \emptyset \nearrow$

A fundamental challenge in integrating multiple VL6180 sensors on a single I2C bus is that each module possesses an identical default factory address (0X29) [10]. Conventional approaches rely on external hardware I2C multiplexers to

route communication, which inadvertently introduces signal latency and increases circuit complexity. To optimize data throughput and streamline the hardware architecture, a dynamic address assignment algorithm was implemented within the microcontroller firmware.

The Arduino Mega 2560 controls the hardware Enable (EN) pins (assigned to GPIO 32, 33, 34, and 35) of the four sensors. During the initialization sequence, all sensors are deliberately placed in a low-power standby state by pulling the EN pins LOW [13], the microcontroller subsequently activates the sensors sequentially. As each sensor wakes up with the default 0X29 address, the firmware immediately establishes communication and reassigns a unique operational address (0X30, 0X31, and 0X32) using a software-driven register adjustment function. The final sensor in the sequence retains the default 0X29 address. This sequential initialization protocol guarantees a conflict-free, high-speed data acquisition loop without the necessity of supplementary multiplexing hardware. The complete firmware source code governing this logic is provided in Appendix A.

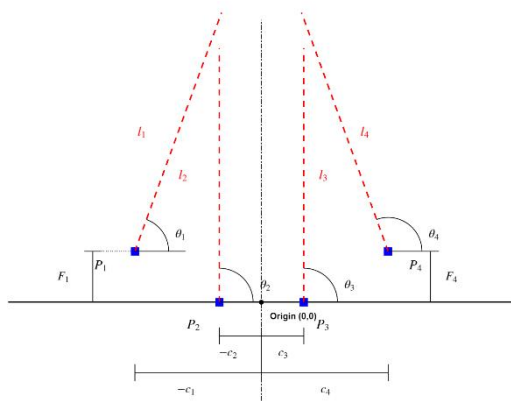


Figure 1: Geometrical coordinate transformation model. The diagram illustrates the spatial arrangement of four sensors (P1, P2, P3, P4) relative to a central origin (0,0). The horizontal distances from the origin to the sensors are denoted as c_1, c_2, c_3, c_4 . The vertical distances from the origin to the points where the sensor's field of view (dashed lines) intersect the vertical axis are denoted as F_1, F_2, F_3, F_4 . The angles between the horizontal axis and the dashed lines are denoted as $\theta_1, \theta_2, \theta_3, \theta_4$. The linear distances from the sensors to the points on the vertical axis are denoted as l_1, l_2, l_3, l_4 .

3.3 Data Acquisition

Real-time linear distance data (l_1, l_2, l_3, l_4) captured by the ToF array is transmitted via serial communication at a baud rate of 115200 to a host computer running Microsoft Excel Data Streamer [14]. To mitigate inherent optoelectronic shot noise and ensure measurement stability, the raw continuous data streams are time-averaged over a defined sampling window before further processing.

The averaged linear distances (l) must be mathematically transformed into a 2D Cartesian spatial coordinate system (x, y) to accurately reconstruct the physical boundary of the target object. This coordinate transformation relies on the known geometric constants of the 3D-printed sensor fixture, specifically the lateral offset from the central origin (c), the vertical baseline height (F), and the specific sensor orientation angle (θ). The geometrical coordinate transformation model is illustrated in Figure 1.

The averaged linear distances (l) must be mathematically transformed into a 2D Cartesian spatial coordinate system (x, y) to accurately reconstruct the physical boundary of the target object. This coordinate transformation relies on the known geometric constants of the 3D-printed sensor fixture, specifically the lateral offset from the central origin (c), the vertical baseline height (F), and the specific sensor orientation angle (θ). The geometrical coordinate transformation model is illustrated in Figure 1.

The mathematical transformation equations for determining the spatial coordinates of the target surface are defined as follows:

$$x_i = \pm c_i + (l_i \cdot \cos \theta_i)$$

$$y_i = \pm F_i + (l_i \cdot \sin \theta_i)$$

By applying these trigonometric functions, the one-dimensional raw range measurements obtained from the four sensors are successfully mapped into four discrete 2D spatial coordinates.

3.4 S—⊙//∠□○
R○□□∅S↗⊙—□↗≡□∅
∠∅∩ S↗∠(P)○
C⊙∠(S)S≡//≡□∠↗≡□∅
A⊙↗□⊙≡↗↗(M)

The final stage of the methodology involves translating the discrete spatial coordinates into a continuous object profile for automated shape classification. Relying on simple linear interpolation (connecting point-to-point) is insufficient, as it fails to account for minor measurement noise and lacks the mathematical foundation to classify complex geometries. Therefore, a polynomial algorithm was employed using MATLAB [15].

The transformed coordinates are processed through a second-order polynomial curve-fitting function, which evaluates the surface boundary using the quadratic formulation [16]:

$$y = ax^2 + bx + c$$

This best-fit continuous profile acts as the core classification algorithm. The leading quadratic coefficient (a) serves as the primary diagnostic indicator, where a negative value ($a < 0$) mathematically classifies the target surface as convex. To empirically validate the algorithm's performance within this study, a cylindrical mug was employed as the sole target object. This specific geometry was selected to demonstrate the system's capability to successfully reconstruct and identify a convex surface profile accurately. This algorithmic approach enables robust, automated shape reconstruction suitable for industrial edge-computing applications without the extensive computational overhead required by vision-based systems.

4. RESULTS AND DISCUSSION

This chapter evaluates the empirical performance of the proposed optoelectronic shape detection algorithm. The primary objective

is to validate the system's ability to accurately reconstruct and classify physical target geometries. Specifically, a convex cylindrical mug using the 4-channel Time-of-Flight (ToF) sensor array and polynomial curve-fitting algorithms.

4.1 D∠↗∠
A□↗—≡(S)≡↗≡□∅ ∠∅∩
C□□⊙∩≡∅∠↗○
T⊙∠∅(S)//□⊙(M)∠↗≡□∅

To test the algorithm's boundary extraction capability, the cylindrical mug was positioned within the active field of view of the 4-channel ToF sensor array. Raw linear distance measurements (l) were successfully captured via the I2C dynamic addressing protocol without any system latency or bus collisions.

The raw distances were then mathematically transformed into 2D Cartesian spatial coordinates (x, y) using the geometric configuration parameters (c, F , and θ) established in the methodology. The distance spatial coordinates recorded from the four channels are presented in Table 4.1.

Table 4.1: Transformed 2D spatial coordinates acquired from the 4-channel ToF sensor array.

Channel ID	Horizontal Axis (x/mm)	Vertical Depth Axis (y/mm)
CH4 (Far left)	-82.14	102.44
CH2 (Inner Left)	-3.50	204.90
CH3 (Inner Right)	3.5	176.33
CH1 (Far Right)	92.18	112.48

The acquired coordinate data demonstrates that the outer sensors (CH1 and CH4) recorded greater depths due to their specific angular mounting, while the inner sensors (CH2 and CH3) captured the nearest protruding surface of the cylindrical mug.

The transformed spatial coordinates were imported into MATLAB to execute the surface reconstruction algorithm. Instead of relying on rudimentary interpolation, a second-order polynomial regression model was applied to the dataset to evaluate the continuous surface boundary of the object.

Executing the polynomial curve-fitting routine (*polyfit*) on the four spatial coordinates yielded the following numerical coefficients:

- Quadratic Coefficients (a): -0.011016
- Linear Coefficient(b): 0.1613
- Constant Coefficient (c): 190.6809

Consequently, the continuous mathematical profile equation governing the target surface is expressed as:

$$y = -0.011016x^2 + 0.1613x + 190.6809$$

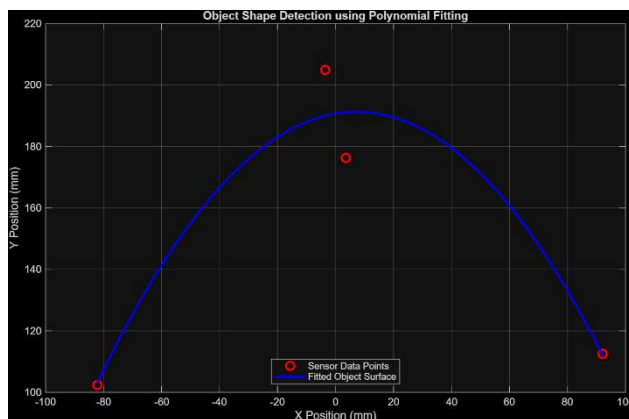


Figure 4.2: Continuous 2D surface profile reconstructed via second-order polynomial curve fitting

4.2 P□L→∅□M≡∠L
C—⊙L□O F≡↗↗↗≡∅/↘ ∠∅∅
S—⊙//∠□O
R□□□∅S↗↗⊙—□↗↗≡□∅

The high degree of fitness across all four sampling points confirms that the low-density spatial data provided by a 4-channel optoelectronic configuration is sufficient to establish a mathematically continuous 2D surface profile.

4.3 S↗∠(P)⊙
C(L)∠(S)S≡//≡□∠↗↗≡□∅
A∅∠(L)→(S)≡(S)

The primary diagnostic indicator for autonomous surface curvature classification within this algorithm lies in the sign and magnitude of the quadratic coefficient, a . In analytical geometry, the sign of a dictates the direction of the parabola's concavity.

The experimentally derived value of $a = -0.011016$ strictly satisfies the condition of $a < 0$, which governs a downward-opening parabola. In the context of the spatial coordinate system used, this mathematical behavior indicates that the central region of the target is physically closer to the sensor baseline than its lateral extremities. This successfully classifies the target object as a convex structure. The result precisely corresponds to the physical geometry of the cylindrical mug utilized in the experiment, validating the algorithm's capability to accurately detect and classify convex object profiles.

4.4 O(P)↗↗□O(L)O□↗↗⊙□∅≡□
C⊙□(S)S↗↗∠(L)∪
M≡/↘⊙∠↗↗≡□∅

A critical limitation in conventional multi-channel ultrasonic and basic infrared sensor arrays is crosstalk where emitted signals scatter and

inadvertently trigger adjacent receivers, resulting in false reading or flattened profile artifacts.

The experimental results validate that the integration of VL6180 ToF technology, coupled with the algorithm filtering, demonstrated robust resilience to optical interference. The stability of the negative a coefficient and the smooth geometric alignment of the coordinate data confirm that stray photons from adjacent emitted were successfully rejected. By eliminating inter-channel crosstalk, the proposed system maintains strict signal integrity across all adjacent channels, allowing for precise quantitative boundary reconstruction despite the compact spacing of the sensor array.

5. CONCLUSION AND FUTURE WORK

5.1 CLOSURE AND FUTURE WORK

This dissertation successfully presented the design, development, and empirical validation of a low-cost, multi-node optoelectronic object shape detection algorithm. By integrating a 4-channel VL6180 Time-of-Flight (ToF) sensor array with an Arduino Mega 2560 microcontroller, the research established a robust framework for real-time spatial profiling that circumvents the computational burden of traditional high-resolution vision systems.

A significant engineering achievement of this study was the implementation of a dynamic I2C address assignment protocol. By controlling the hardware Enable (EN) pins via the microcontroller firmware, the system sequentially initialized and reassigned unique operational address to identical sensors. This software-driven approach eliminated the need for external I2C multiplexers, thereby reducing circuit complexity, minimizing hardware latency, and ensuring a conflict-free data acquisition loop.

The core shape detection algorithm effectively transformed one-dimensional raw distance measurements into a 2D Cartesian spatial coordinate system using specific geometric

constants. Rather than relying on simple linear interpolation, the system employed a second-order polynomial curve-fitting algorithm ($y = ax^2 + bx + c$) in MATLAB to reconstruct the continuous surface profile. The experimental validation, conducted using a cylindrical mug, yielded a definitive negative quadratic coefficient ($a = -0.011016$). This mathematical parameter successfully and autonomously classified the target as convex geometric structure.

Furthermore, the integration of highly directional ToF technology completely mitigated the severe inter-channel crosstalk and target reflectance biases that typically plague conventional ultrasonic and basic infrared sensor arrays [4]-[6]. The findings confirm that the proposed 4-channel optoelectronic algorithmic array provides a highly pragmatic, accurate, and color-independent solution for object shape recognition, ideally suited for edge-node industrial automation and robotic proximity sensing.

5.2 FUTURE WORK

While the proposed system demonstrated high accuracy and robust performance, certain limitations provide avenues for future research and development

Expanded Geometric Classification: The empirical validation in this study was specifically constrained to a convex target (a cylindrical mug) to prove the fundamental concept. Future work should expand the testing dictionary to include flat surfaces, concave profiles, and complex asymmetric geometries to fully establish the algorithm's classification boundaries.

Surface Material Constraints: Although ToF sensors are largely color-independent, extreme surface materials such as highly transparent glass or highly specular reflective metals can disrupt the optical photon return path. Future studies should evaluate the sensor array's performance across varying surface opacities and specularities.

Increased Sensor Density and Machine Learning: To achieve higher spatial resolution, the array could be expanded to include more ToF sensor nodes (e.g., 8-channel or 16-channel array). Coupled with a higher density of spatial data, implemented lightweight Machine Learning (ML) classifiers on the edge microcontroller could transition the system from basic geometric curve fitting to advanced real-time object recognition.

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 12C $A \textcircled{S} \textcircled{S} = \cancel{P} \cancel{Q} \textcircled{M} \odot \emptyset \nearrow$

```

/*
 * Dissertation Project: 4-Channel VL6180 ToF Sensor Array
 * Microcontroller: Arduino Mega 2560
 * Description: Dynamic I2C address assignment using hardware Enable (EN) pins
 * to bypass external I2C multiplexers.
 */

#include <Wire.h>
#include <Adafruit_VL6180X.h> // Standard Adafruit library name for VL6180 series

// Define hardware Enable (EN) pins on Arduino Mega 2560
#define EN_PIN_1 32
#define EN_PIN_2 33
#define EN_PIN_3 34
#define EN_PIN 4 35

```

```
// Define new I2C operational addresses
#define ADDRESS_1 0x30
#define ADDRESS_2 0x31
#define ADDRESS_3 0x32
// Sensor 4 will retain the default address: 0x29

Adafruit_VL6180X sensor1 = Adafruit_VL6180X();
Adafruit_VL6180X sensor2 = Adafruit_VL6180X();
Adafruit_VL6180X sensor3 = Adafruit_VL6180X();
Adafruit_VL6180X sensor4 = Adafruit_VL6180X();

void setup() {
  Serial.begin(115200);
  Wire.begin();

  // Step 1: Set all EN pins as OUTPUT and pull LOW to put sensors in standby
  pinMode(EN_PIN_1, OUTPUT);
  pinMode(EN_PIN_2, OUTPUT);
  pinMode(EN_PIN_3, OUTPUT);
  pinMode(EN_PIN_4, OUTPUT);

  digitalWrite(EN_PIN_1, LOW);
  digitalWrite(EN_PIN_2, LOW);
  digitalWrite(EN_PIN_3, LOW);
  digitalWrite(EN_PIN_4, LOW);
  delay(10);

  // Step 2: Initialize Sensor 1 and reassign address
  digitalWrite(EN_PIN_1, HIGH);
  delay(50);
  sensor1.begin();
  sensor1.setAddress(ADDRESS_1);

  // Step 3: Initialize Sensor 2 and reassign address
  digitalWrite(EN_PIN_2, HIGH);
  delay(50);
  sensor2.begin();
  sensor2.setAddress(ADDRESS_2);

  // Step 4: Initialize Sensor 3 and reassign address
  digitalWrite(EN_PIN_3, HIGH);
  delay(50);
  sensor3.begin();
  sensor3.setAddress(ADDRESS_3);

  // Step 5: Initialize Sensor 4 (Retains default 0x29)
```

```

digitalWrite(EN_PIN_4, HIGH);
delay(50);
sensor4.begin();

Serial.println("All ToF sensors initialized successfully.");
}

void loop() {
  // Read distance from all 4 sensors
  uint8_t l1 = sensor1.readRange();
  uint8_t l2 = sensor2.readRange();
  uint8_t l3 = sensor3.readRange();
  uint8_t l4 = sensor4.readRange();

  // Transmit raw distance data via serial for Excel Data Streamer
  Serial.print(l1); Serial.print(",");
  Serial.print(l2); Serial.print(",");
  Serial.print(l3); Serial.print(",");
  Serial.println(l4);

  delay(100); // 10Hz sampling rate
}

```

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This appendix provides the MATLAB script utilizing for processing the transformed 2D spatial calculates the mathematical coefficients to autonomously classify the target's geometric profile and generates the corresponding visual surface plots.

```

% -----
% MATLAB Code: Polynomial Curve Fitting & Surface Reconstruction
% Target: Cylindrical Mug (Convex Surface Validation)
% -----

% Step 1: Input transformed 2D spatial coordinates (x, y) in mm
% Data extracted from the 4-channel ToF sensor array
x_coords = [-82.14, -3.50, 3.50, 92.18];
y_coords = [102.44, 204.90, 176.33, 112.48];

% Step 2: Perform 2nd-order polynomial curve fitting (y = ax^2 + bx + c)
degree = 2;
coefficients = polyfit(x_coords, y_coords, degree);

% Extract individual mathematical coefficients
a = coefficients(1); % Quadratic coefficient (Shape classifier)

```

```

b = coefficients(2); % Linear coefficient
c = coefficients(3); % Constant coefficient

% Display coefficients in the command window for diagnostic review
fprintf('Quadratic Coefficient (a): %f\n', a);
fprintf('Linear Coefficient (b): %f\n', b);
fprintf('Constant Coefficient (c): %f\n', c);

% Step 3: Shape Classification Logic
if a < 0
    fprintf('Diagnostic Result: Target surface classified as CONVEX.\n');
else
    fprintf('Diagnostic Result: Target surface classified as CONCAVE/FLAT.\n');
end

% Step 4: Generate continuous profile points for plotting
x_fit = linspace(min(x_coords)-10, max(x_coords)+10, 100);
y_fit = polyval(coefficients, x_fit);

% Step 5: Plot the empirical data points and the reconstructed continuous surface
figure('Color', 'w'); % Set background to white for academic report printing
plot(x_fit, y_fit, 'b-', 'LineWidth', 2); % Reconstructed boundary
hold on;
plot(x_coords, y_coords, 'ro', 'MarkerSize', 8, 'LineWidth', 2); % Raw data points

% Formatting the graphical output
title('Object Shape Detection using Polynomial Fitting', 'FontSize', 14);
xlabel('X Position (mm)', 'FontSize', 12);
ylabel('Y Position (mm)', 'FontSize', 12);
legend('Fitted Object Surface', 'Sensor Data Points', 'Location', 'south');
grid on;
hold off;

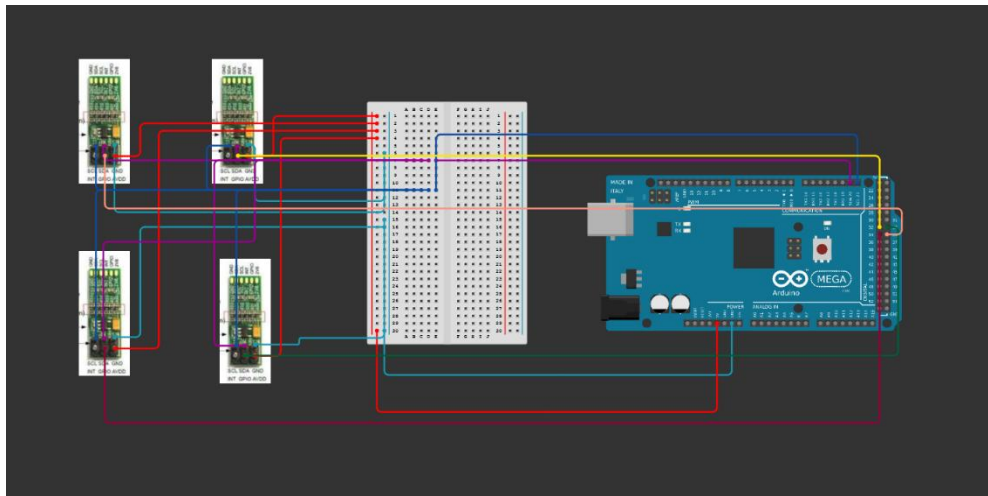
```


Experimental Hardware and Data Acquisition Setup

This section provides comprehensive visual documentation of the physical experimental setup, hardware circuit architecture, and data acquisition interfaces utilized to empirically validate the 4-channel optoelectronic shape detection system.

C.1 Optoelectronic Sensor Details

The foundational measurement unit for this research is the VL6180 Time-of-Flight (ToF) proximity module. As depicted in Figure C.1, the sensor is integrated into a breakout board featuring a 2.8V regulator and a 6-pin interface. This configuration is critical for establishing stable I2C communication and enabling the dynamic address reassignment protocol via the microcontroller's hardware Enable (EN) pins.



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C.2

Integrated Hardware Circuitry

The

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complete eletromechanical architecture of system is

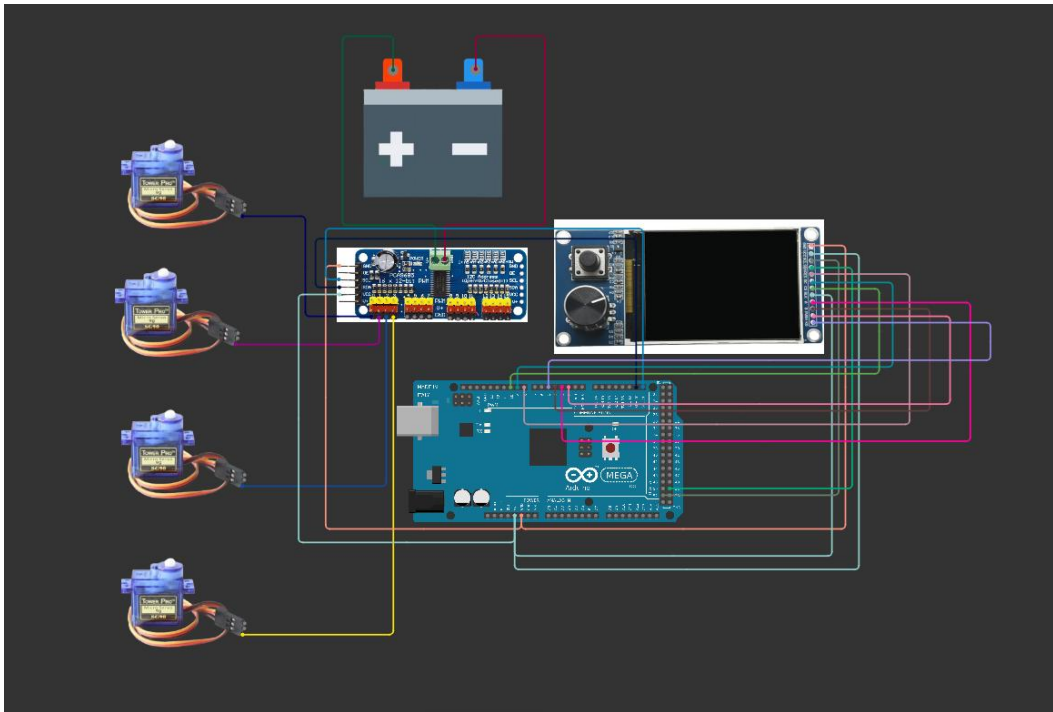
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illustrated in Figure C.2. These comprehensive schematics details the integration of three primary subsystems with the Arduino Mega 2560 microcontroller:

Sensor Array: The I2C bus routing (SDA/SCL) and discrete hardware Enable (EN) connections (Digital Pins 32-35) required for the dynamic addressing of the four VL6180 modules.

Actuation Systems: A PCA9685 16-channel, 16-bit PWM controller powered by an independent external DC supply. This ensures stable, synchronous operation of the four micro-servo motos used for angular sensor positioning (θ).

Local User Interface: A TFT LCD module equipped with a 5-way joystick. The display communicates via the hardware SPI bus (Pins 50-52), while the joystick inputs (configured as Active LOW) are routed to standard digital pins to facilitate real-time system calibration and manual mechanism control.



C.3

Physical

Implementation

To conduct empirical shape-detection trials, the integrated hardware was deployed on a custom 3D-printed testing rig. As shown in Figure C.3, the four sensors are mounted on adjustable servo brackets along a linear guide mechanism. This physical setup allowed for precise spatial alignment when scanning the target object (a cylindrical mug) to validate the polynomial curve-fitting algorithm.

C.4 Data Acquisition Interface

